

LOSO Experiment C_left_Istm

LSTM only · Full sensors, LEFT hand only, no Nothing

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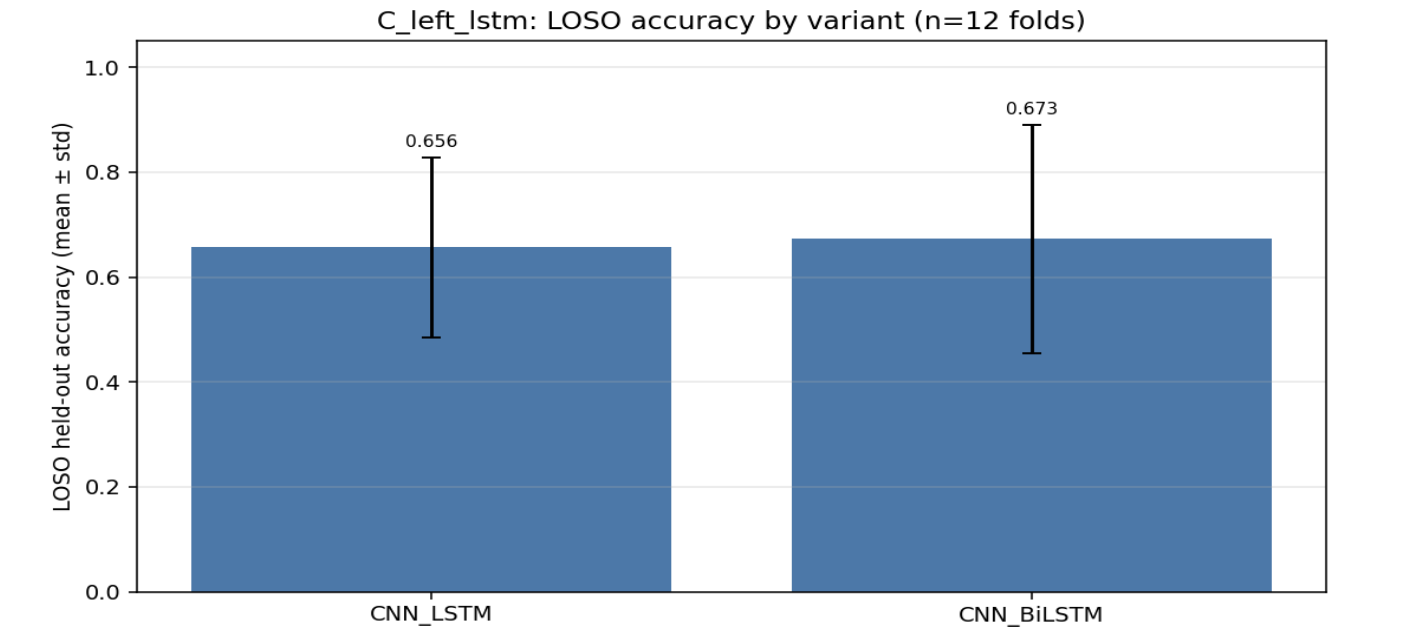
Experiment configuration

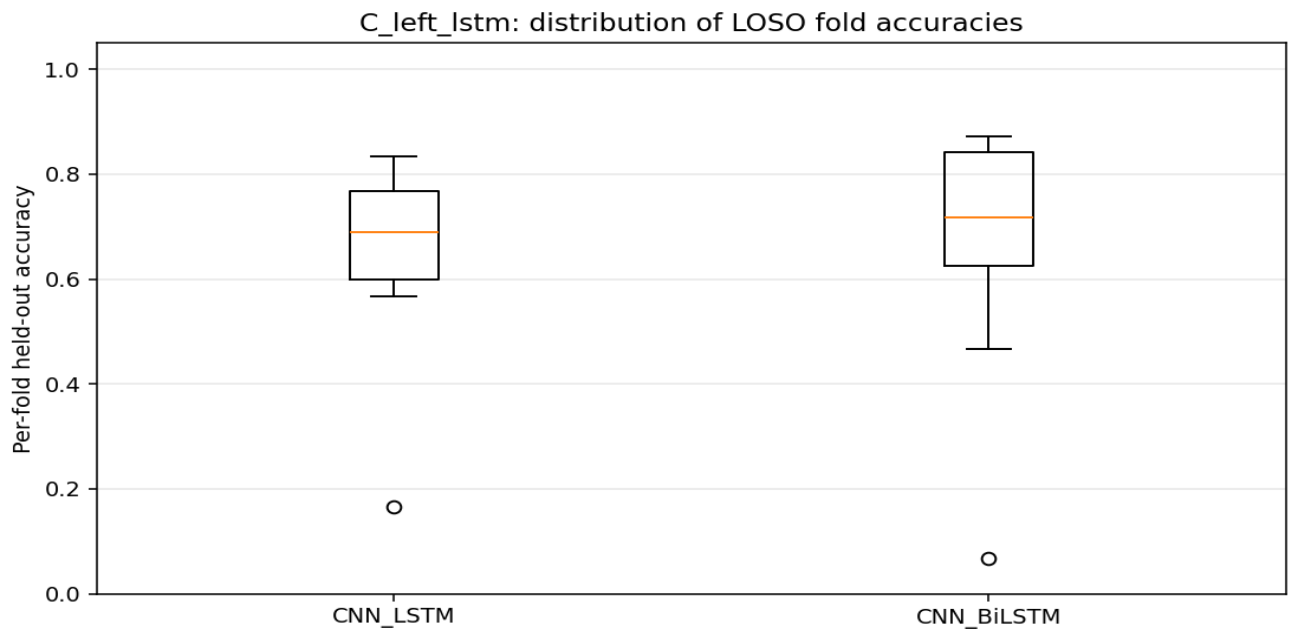
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	left
Sensor channels	88
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

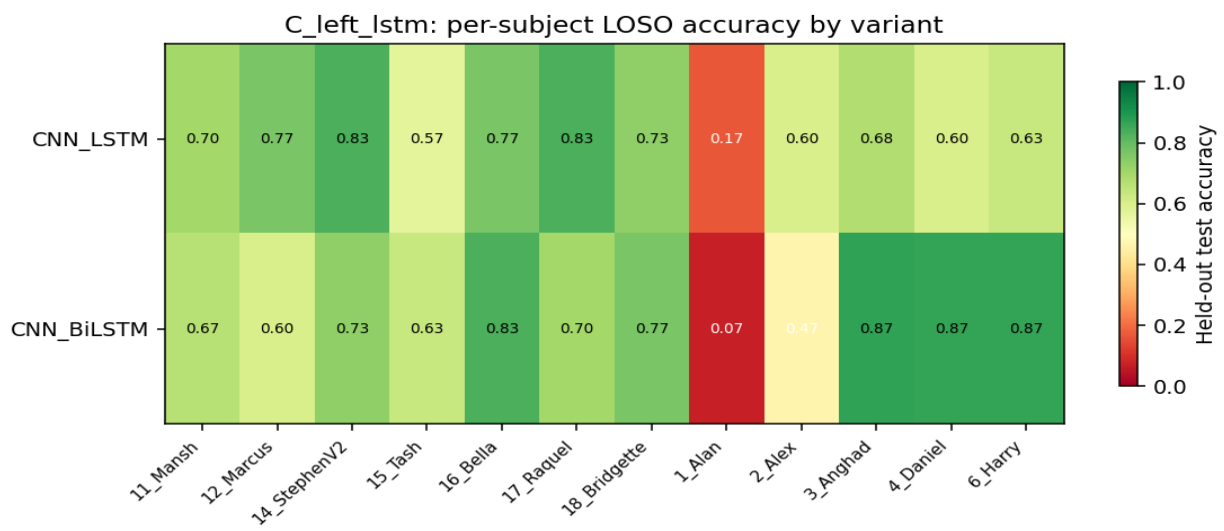
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	57,126	0.6565	0.1708	0.1667	0.8333	0.7260
CNN_BiLSTM	94,246	0.6726	0.2180	0.0667	0.8710	0.7815

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.700	0.667
12_Marcus	331	0.767	0.600
14_StephenV2	331	0.833	0.733
15_Tash	331	0.567	0.633
16_Bella	331	0.767	0.833
17_Raquel	331	0.833	0.700
18_Bridgette	331	0.733	0.767
1_Alan	331	0.167	0.067
2_Alex	331	0.600	0.467
3_Anghad	330	0.677	0.871
4_Daniel	331	0.600	0.867
6_Harry	331	0.633	0.867

Variant: CNN_LSTM

Params: 57,126

LOSO mean accuracy: 0.6565 ± 0.1708 (min 0.1667, max 0.8333)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	1/5	5/5	5/5	5/5	5/5	0/5	0.700
12_Marcus	2/5	4/5	5/5	5/5	5/5	2/5	0.767
14_StephenV2	5/5	5/5	5/5	0/5	5/5	5/5	0.833
15_Tash	5/5	5/5	5/5	0/5	2/5	0/5	0.567
16_Bella	5/5	5/5	5/5	0/5	5/5	3/5	0.767
17_Raquel	5/5	5/5	5/5	0/5	5/5	5/5	0.833
18_Bridgette	5/5	5/5	3/5	0/5	5/5	4/5	0.733
1_Alan	5/5	0/5	0/5	0/5	0/5	0/5	0.167
2_Alex	0/5	3/5	5/5	5/5	5/5	0/5	0.600
3_Anghad	0/5	5/5	5/5	0/5	6/6	5/5	0.677
4_Daniel	5/5	5/5	5/5	0/5	1/5	2/5	0.600
6_Harry	0/5	5/5	5/5	0/5	4/5	5/5	0.633
MEAN	0.633	0.867	0.883	0.250	0.783	0.517	0.656

Variant: CNN_BiLSTM

Params: 94,246

LOSO mean accuracy: 0.6726 ± 0.2180 (min 0.0667, max 0.8710)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	0/5	5/5	0/5	0.667
12_Marcus	3/5	0/5	5/5	5/5	5/5	0/5	0.600
14_StephenV2	2/5	5/5	5/5	0/5	5/5	5/5	0.733
15_Tash	1/5	5/5	5/5	1/5	2/5	5/5	0.633
16_Bella	5/5	5/5	5/5	5/5	4/5	1/5	0.833
17_Raquel	5/5	5/5	5/5	0/5	5/5	1/5	0.700
18_Bridgette	0/5	5/5	3/5	5/5	5/5	5/5	0.767
1_Alan	2/5	0/5	0/5	0/5	0/5	0/5	0.067
2_Alex	0/5	4/5	5/5	0/5	5/5	0/5	0.467
3_Anghad	1/5	5/5	5/5	5/5	6/6	5/5	0.871
4_Daniel	5/5	5/5	5/5	5/5	1/5	5/5	0.867
6_Harry	5/5	5/5	5/5	2/5	4/5	5/5	0.867
MEAN	0.567	0.817	0.883	0.467	0.767	0.533	0.673